University of Glasgow | School of Computing Science

Honours Individual Project Dissertation

PUMP AND DUMP: IDENTIFYING CRYPTO-MARKET MANIPULATION ON SOCIAL MEDIA

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March 24, 2023

Abstract

Since the introduction of Bitcoin in 2009 with its revolutionary blockchain architecture, There are many emerging cryptocurrencies, all of which have special features that set it apart from the others. However, due to high volatile nature of cryptocurrencies, it can be manipulated by individual or group of people in social media order to gain profit. This project aims to address this issue by devising a machine learning model in order to determine a pump and dump is happening by utilizing both financial and social data. This project will then compare different learning models in order to find the best models. Finally, an attempt to predict where the pump and dump is ended was also made during this project.

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1 Introduction

1.1 Motivation

Before money was invented, people around the world used to have a barter system, whereby goods or services were traded in exchange for other goods or services. Fast forward in a few centuries, a more efficient medium of exchange has made known as currency. Traditional currency is commonly utilised as means of exchanging goods and service in the world. Currency used to be backed by gold, which caused the Currency supply to be constricted. However, during mid 20th century, the concept of having reserve amount of gold for the currency was removed and instead new concept was introduced in which has given different nations a power to control the supply of their own currency, it is called fiat currency. Fiat currency relied on government sovereignty, it means the value of fiat money is derived from relationship between the supply and demand and the stability of issuing government. Most countries use fiat currency including US dollar, the euro, and any major global currencies.

In 2009, another type of currency was introduced by Satoshi Nakamura, known as Bitcoin. Bitcoin is a cryptocurrency, in sense that the currency is controlled with cryptography for its creations and transaction rather than central authority. Bitcoin use a network under public ledger called the blockchain, and ledger are secured with consensus mechanism by proof-of-work (POW). The ledger contains every transaction ever recorded allowing user's to verify the validity of each transaction. Since the rise of Bitcoin, there are numerous cryptocurrencies emerging, with each offering unique features that distinguish them from one another.

The difference in cryptocurrency and fiat is that cryptocurrency ensure anonymity while also offering transparency at the same time. Cryptocurrency is also decentralized currencies, meaning cryptocurrency has no involvement with government. Another difference can be look at the distribution of fiat and cryptocurrency. while fiat currency does not have limited on supply, most cryptocurrency are designed to have a predetermined and limited supply of coins that cannot be increased or decreased. Lastly, fiat currency can be stored in both physical and digital forms, whereas cryptocurrency can only be stored in digital forms.

However, despite the revolutionary blockchain technology used for cryptocurrencies, it appears that most of cryptocurrencies are unstable. While some cryptocurrencies can be categorised as "stablecoins"—to put simply a cryptocurrencies can be determined as stablecoin if its backed by fiat currency or commodities. Majority of cryptocurrency is not backed by anything. Therefore, most of the cryptocurrencies price is determined based on their supply and demand and people's speculative assumptions, resulting in high volatility.

Because of this, many individuals or group tried manipulate the cryptocurrency space for their personal gain. Furthermore the anonymity offered in cryptocurrency space and its decentralised nature is often abused by bad actors in order to carry out fraudulent scheme, such as pump and dump scheme. Some times these bad actors also use social signals and create a sense of urgency in order to lure unaware investors.

1.2 Aims

The aim of this project is to develop a machine learning algorithm that able to leverage both cryptocurrency market and social data in order to predict pump and dump event. a pump and dump scheme is a type of financial fraud where the individual or a group operates to artificially inflate the price of a security and then sell it by first owning the said security and then advertising false information to convince unaware investors to drive the price up and finally sell his security at the profited margin. To achieve this, the machine learning algorithm will be trained in dataset that contains historical market price and social media information, in this case data from Twitter will be used as our social media data. The algorithm will then analyse and learn from the pattern to identify crucial factors that may precede to pump and dump event.

2 Background

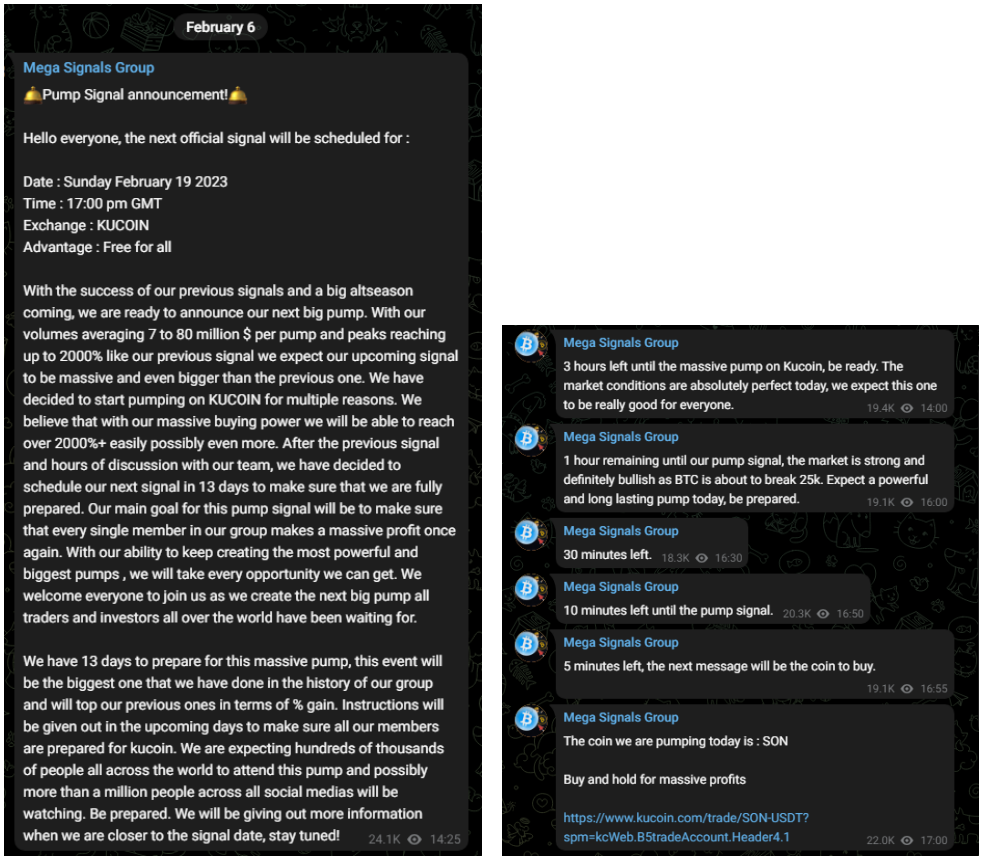
2.1 Pump and Dump

Kyle and Viswanathan (2008) define a pump and dump scheme is a type of financial fraud where the individual or a group operates to artificially inflate the price of a security and then sell it by first owning the said security and then advertising false information to convince unaware investors to drive the price up and finally sell his security at the profited margin. One of the famous cases of pump and dump dates back to the late 1920s. A group known as "The Radio Pool" tried to artificially inflate the stock price of a major American electronics company Radio Corp. of America, better known as RCA. The group circulate false information about the stock through, radio, tabloids, and word of mouth (Morgia et al. 2020). They managed to pump the RCA stock price to an astronomical value of $549 and then dump the shares, causing the price to collapse just under $10 in 1929. Many people who bought the shares lost all their savings because of this. Surprisingly, The Radio Pool practice of colluding to manipulate the stock price artificially was legal before the US Securities and Exchange Commission (SEC) outlawed it in the 1930s.

In cryptocurrency era, pump and dump schemes are more prevalent than ever. This is due to the nature of cryptocurrency being highly accessible to the masses, having high volatility, and lack of regulation due to its decentralized aspect making it easy for bad actors to manipulate the crypto market for profit.

Pump and dump that executed In crypto spaces are done by one or multiple self-organized groups of people on the internet. Therefore, anyone can join the groups. These groups use Telegram or Discord servers to carry out their pump and dump operations (Xu and Livshits 2019). Despite the accessibility of these pump and dump groups to the general public, a hierarchical structure within the group persists, higher hierarchy means the user is able to gain insight on which crypto is going to get pumped and the timing of the pump earlier than other group members (Morgia et al. 2020). This way users with higher hierarchy can gain even more profit from the pump event. The hierarchical structure of some of these groups is based on an affiliation system or VIP system, whereby individuals climb the hierarchy by inviting more people to join the group or pay a fee to gain higher hierarchical order in the group.

The typical process of a pump and dump can be illustrated the by examining a specific pump and dump event occurred on 19th February 2023. On 6th February 2023, the admin of *Mega Signals Group* announced their upcoming pump, detailing its date, time, and exchange (figure 2.1a). The admin also mentioned that the potential pump would be "free for all," meaning it would be accessible to everyone who were aware of this behaviour. Along with that, the admin features the history of their last pump and its result to further convince the group members to participate in the next pump. For the next few days, the admin would send a message as remainder before the pump. Occasionally, the admin may send instructions on how to conduct trading on the target exchange. During the day of pump the admin will send a message in shorter interval. Right when at 17:00 UTC + 0, the admin revealed the target coin to be SON, corresponding to Souni token(figure 2.1b). After the pump start, admin will send a tweet or a news and encourage all members to share the tweet or news to social media about the rising cryptocurrency price. This



(a) The pump announcement *(b) pump announcement on the scheduled day*

Figure 2.1: *Pump announcement on Mega Signals Group (a) shows the pump announcement by admin, detailed with date, time, and target exchange. (b) As the pump event draws closer, the admin will send the reminders in shorter intervals. Then, on the scheduled pump time of , the admin will reveal the target crypto to be pumped.*

is done to create a FOMO (fear of Missing Out) of an unmissable investment opportunity that will draw investors from outside the group.

Price and volume of the target coin could highly fluctuate in short amount of time, leaving unaware buyers with their share of useless coin at the end of pump (Kamps and Kleinberg 2018). Figure 2.2 shows the market activities between SON token and USDT during the pump on 19th February 2023. The top illustration is a candlestick chart, a chart widely used in both traditional stock markets and cryptocurrency markets. A green rectangle indicates higher closing price while red rectangle indicate the opposite. vertices on vertical bars represents a highest and lowest coin price at particular time point. The bottom part of histogram records the coin's volume traded with respect to time. Before the pump started, SON token trading activity remains minimal and when the pump is announced it immediately raise up the coin's price and volume, only to drop back to original price few hours later.

In the stock exchange, pump and dumps primarily involve penny stock—that is a stock that priced $5 or lower per shared according to SEC. Penny stock manipulation that is done by using deceptive social media campaigns to amass profits and can be considered as a criminal activity (Xu and Livshits 2019). However, because many cryptocurrencies cannot be categorized as investments (Li and Mann 2018), specific security laws may be dubious and at the moment,

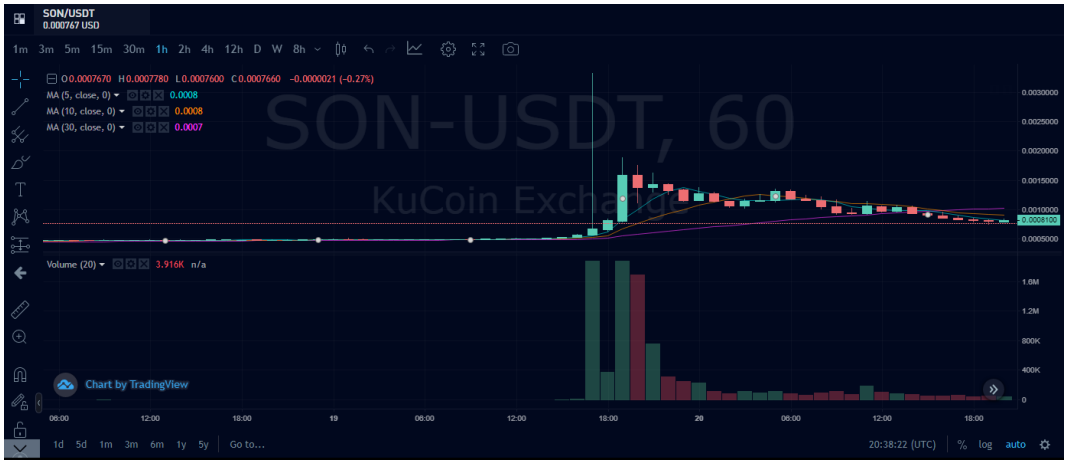


Figure 2.2: SON's coin open, high, low, close, volume (OHLCV) graph before the pump, during the pump on 19th February 2023, and after the pump. screenshot taken from kucoin.com.

regulations and laws governing the cryptocurrency space are either inadequate or non-existent (Li et al. 2021). As pump and dump group admins announce the pump, they intentionally deceived unaware investors for their own financial gain and possibly their own group members. Therefore, under US Federal Trade Commission (FTC) Act and UK Consumer Protection from Unfair Trading Regulations 2008 Act, these admins might be committing financial fraud through false advertising.

Organizing pump and dump schemes and aiming to gain financial profit from manipulating cryptocurrency by these admins are indeed unethical. Nevertheless, the group members are also responsible, since their actions empowered and augmented such action to happen (Xu and Livshits 2019). ironically, these groups members become the victim of their admins' scheme by their own volition.

2.2 Social Media Manipulation

According to Cambridge dictionary, *manipulation* can be define as "to control someone or something to your own advantage, often unfairly or dishonestly". The dictionary defines *social media* as "forms of media that allow people to communicate and share information using the internet or mobile phones". Therefore, social media manipulation can be defined as "an act of using social media unfairly or dishonestly to control individual or groups". The purpose of social media manipulation is that intended to deceive and mislead people, causing harm, or for political and financial gain.

Over several years, social media platform such as Facebook, Twitter, Instagram, and TikTok have amassed millions of users, revolutionizing our means to communicate with others by providing instantaneous access to the exchange of written, visual, and video messages and/or post with individuals both known and unknown from around the world. Although social media was intended to facilitate communication among users by breaking time and location barrier, it has evolved into a place where people can also share and obtain information. According to research done by Pew Research Center on 31st August 2020 until 7th September 2020, 53% of American adults use social media as their source of information "often" or "sometimes" (Shearer et al. 2022). Facebook is become social media of choice for American adults as their source of news with 36% American responded that they regularly get news on Facebook. Followed by YouTube and Twitter with 23% and 15% of US adults respectively reported getting news on these sites.

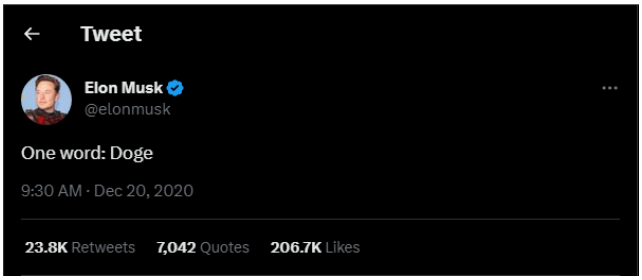


Figure 2.3: *Elon Musk's tweet on 20th December 2020.*

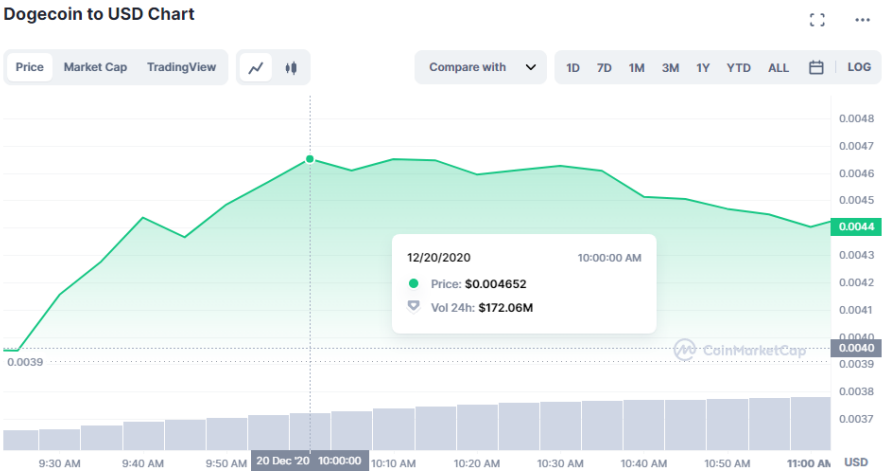


Figure 2.4: *Doge coin price after Elon Musk tweet. Screenshot captured from coinmarketcap.com*

With the evolution of social media, from a platform to shares personal thoughts, cat images, and memes, it has become a new source for people to obtain news. This has led to the potential for bad actors to misuse of disseminating information to manipulate the masses through social media.

The rise of social media has led to the emergence of many personalities and so-called financial groups or communities claiming to be experts in the financial sector, which has had an impact on the industry. While some of these personalities and communities may come from a proper background, have more experience, and be well-informed, a large number of internet personalities are entirely based on speculation, are ill-informed, and could potentially mislead their audience (Delfabbro et al. 2021).

Many people seem turn a blind eye to these so-called "expert". One such cases can be illustrated by looking at subreddit r/WallStreetBets (WSB). WSB is a Reddit forum where members post their gains and losses from trading, their trading strategy, and memes related to financial market. The difference between WSB from other finance-related subreddit lies from the trading strategy used of majority of the subreddit members. Most of the subreddit member's strategy involves betting all of their fund on stocks—essentially gambling their money away. During 2021, underrated stocks as GameStop (GME) were targeted by WSB members to initiate a short squeeze of institutional investors who had bet against the stock's decline. The event is described in a brief and simplistic fashion in Chohan (2021). A short squeeze occurs when investors who have shorted a stock are forced to buy shares to cover up their position, causing the stock price to rise. With their coordinated effort, WSB is able to hike up GME stock price to $500 dollars on 28th January 2021. Because WSB coordinated movement, institutional investors lost millions of dollar because they had cover up their positions by buying back shares at much higher price.

Social media manipulation around cryptocurrency spaces cannot be overlooked either.The highly volatile and speculative nature of many cryptocurrencies makes it easier for individuals or group to manipulate others through social media. An instance of this can be seen by glancing at Elon Musk tweet on 20th December 2020. on 09:30 UTC+0, Elon Musk tweeted "One word: Doge" (figure 2.3), referring to cryptocurrency named Dogecoin. A cryptocurrency that started as joke, founded in 2013 by Jackson Palmer and Shibetoshi Nakamoto (Dogecoin). The tweet get many attention from people with more than 200.000 people liked the tweet and over 23.000 people retweeted Elon's tweet. More importantly, Elon's tweet caused the price of Dogecoin to increase around 12%, while also increasing Dogecoin volume traded by 79.2% in just 30 minutes after the tweet (figure 2.4). While it is unclear what prompted Elon to publish such a tweet, it is evident that his tweet had a significant impact on Dogecoin prices.

Regardless of whether social media manipulation targeted is operated with the intention to challenge institutional investors or create sense of urgency within a public domain, its ability to influence individuals to unconsciously cave in to its ploy, remains a cause for concern. This holds true regardless of whether such manipulation is executed deliberately or inadvertently. The vulnerability of individuals to fall prey to social media manipulation can be attributed to two factors, namely, **herd mentality** and **the fear of missing out (FOMO)**.

Herding is the phenomenon of individuals choosing to follow others and imitate group behaviours rather than making independent decisions based on their own, individual understanding (Baddeley 2010). Keynes et al. (1930) perceived herding as a reaction to uncertainty and people's perceptions of one's own ignorance: those may follow the crowd because they assume the rest of the crowd ought to be more informed. This can generate instability and within financial markets, herding is a key factor in generating speculative bubbles.

In WSB GameStop saga, Numerous individuals took part even tough considerable proportion of them lacked understanding on how stock market work. They merely involved in the movement due to the participation of others. several studies also linked herd mentality exist within stock market investors. Dang and Lin (2016) found that investors will discard their own reasoning and abruptly follow other investors when market experienced large price movement. Therefore this act will pushed the market price and lowering the market return. Beside lowering the market return, herding could lead to a market bubble—a period when market is filled with speculation whereas negative herding would cause the bubble to burst, leading to market crash (Shantha 2018).

On the other hand, fear of missing out (FOMO) can be defined as "the feeling of unease that others may be enjoying rewarding experiences that one is not part of" (Przybylski et al. 2013). Przybylski et al. (2013) also stated that FOMO is associated with lower life satisfaction and higher social media engagement. Furthermore FOMO is also correlated with problematic smartphone use (Elhai et al. 2016). This means traders who use social media on regular basis will experienced FOMO more frequently than the ones who rarely use it.

FOMO is becoming one of the strongest psychological factor that influence crypto traders (Delfabbro et al. 2021). They may experience feelings of remorse for having refrained from acquiring a coin whose value has since appreciated, despite having previously considered its purchase. Feelings frustration because they could not be a part of the movement. Arguably the single most troublesome part of the circumstance is when the trader see a price of certain crypto going up, and they feel obligated to be a part of the action, they bought the coins at the all time peak only to witness the price drop downs because the coin had been pumped and dumped. In Dogecoin case, many people started buy the Dogecoin right after Elon's tweet. Their desire to be a part of Dogecoin price up movement under Elon Musk's tweet. in conclusion, it is important for traders to remember that investing in cryptocurrencies involves risk, and making well-informed decision based on market analysis rather than emotional and social impulse can help prevent regret and loss in the long run.

2.3 Related Works

Several researches have investigated the complex relation between social media and financial market. Nguyen et al. (2015) suggested a model that incorporate sentiment analysis gathered from social media could enhance stock market prediction better than model that only use historical market data. Cryptocurrencies market also exhibited similar pattern (Phillips and Gorse 2017). Shen et al. (2019) found that the number of tweets on Twitter can significantly predict Bitcoin future realized volatility and its volume trading. Furthermore Garcia and Schweitzer (2015) extracted Twitter valence and polarisation as a derived feature managed to improve trading tactics for Bitcoin. Glenski et al. (2019) constructed a machine learning models by combining 2 social media signals from Github and Reddit and able to enhance price forecast for Bitcoin, Ethereum, and Monero. By analysing sentiment extracted from associated tweet with specific altcoins, Steinert and Herff (2018) managed to shown certain level of proficiency in predicting shot-term returns. For detecting pump and dump scheme, Morgia et al. (2020) managed to devise model that able to detect pump and dump in 5 seconds with accuracy of 92.4% using derived featured from open, high, low, close, volume (OHLCV) data. Nghiem et al. (2021) tried to use unsupervised learning model to detect target coin for pump and dump scheme and its effectiveness managed to construct a models that able to predicts coins' prices regardless there's a pump and dump or not. However, they also found that social signals generally do not appear to enhance the pump price prediction.

Researchers have attempted to predict cryptocurrency price utilising different set of features for various purposes. The previous work is expanded by utilising more granular financial data and social data in hope to gain a better insight of pump and dump event. Additionally, in this project an attempt is made to determine whether social signal in finer details could perform better and to predict when the pump and dump event ended. The project conducted offers valuable insights into the possible role social media data at different stages of a typical pump and dump scheme in the cryptocurrency market.

3 Requirements

Requirements in this project are identified during the initial stage. They were later revised, revisited, and updated updated in response to feedback from the project supervisor.

3.1 User Stories

These user stories are made to help identify the requirements needed for the project. They served as a representation of how the end user would use the product and illustrate the potential usage of the product.

James is a Biology student in University of Glasgow. He took a student loan in order to finance his tuition fee. He worked part-time as store clerk in order to paid his student loan. However, during COVID-19 pandemic, he cannot work as part-timer because the shop where he work is closed. Because of this, he is trying to find a new way to make money to pay off his student debt. His friend suggesting that he should be investing in cryptocurrencies. However, since he does not have knowledge in cryptocurrency market and he seen many news on crypto pump and dump cases, he is looking for a solution where he can get some information whether a cryptocurrency has been pumped or not.

Michael is a 29 years old unemployed man that claims to be a day trader in crypto space. He sells and buys crypto and profited from it based on short-term movement in prices using his savings. Because the nature of day trading in crypto market, sometimes Michael fell for a pump and dump scheme. He thought he could gain profit from sudden movement of cryptocurrencies only to realized he left with useless coin at the end. He thought it would be nice for him and other crypto day traders if there exist a website or a system market data in real time to find out whether a cryptocurrency is being pump or dump.

John is a 17 years old high school student. He has interest in cryptocurrency market because his older brother introduced it to him. While he is looking on the crypto exchange to decide which coin he should invest his money into, he saw a price increase of 200% from a coin despite minimal trading activity over the past two weeks. He tried to look for some clues in Twitter, he found many people were tweeting about the same coin with phrases such as "to the moon!", "Elon will notice this movement", "join the hype train now!", etc. while the coin price keep increasing, John was left confused about whether he should invest his money in the coin or wait for a few minutes in case the movement was just a pump and dump scheme. he thought it would be nice if there is exist a system that can combine social data and market data to determine whether a tweet might be related to pump and dump scheme.

3.2 Requirements

The requirements are partitioned by its significance using MoSCoW approach (Iiba and Analysis 2009)—that is separating list of requirements into different categories, such as "Must Have", "Should Have", "Could Have", and "Won't Have".

3.2.1 Must Have

These requirements are critical for the successful of the project. they provide as a base in which extra feature can be added later on during the project

- **M.1** Query the cryptocurrency —user can query a cryptocurrency and then returns back the model predictions.
- **M.2** Models creation—construct a supervised machine learning model that capable to predict whether a pump and dump is happening using minute data.
- **M.3** Presenting the result—present the guesses made by the machine learning model in a way that is easily understandable to general public.

3.2.2 Should Have

These requirements are assigned into a high degree of importance, but they are not necessary to for the product to be able to accomplish its main tasks.

- **S.1** display the market historical data—the system would show the price and volume during the pump and dump event.
- **S.2** present the tweets that lead to the pump and dump—this is done in order to highlight the tweets that possibly leads to the pump and dump event.

3.2.3 Could Have

These requirements are desirable, because it will improve user satisfaction by augmenting the set of feature of the product. While not essential, they would be a valuable addition if time permits.

- **C.1** shows crypto coin's past pump and dump event—this is done in order to show how many pump and dump event happened in a cryptocurrency and to generate user awareness of past pump and dump, thus they can make judgement next time they want to buy cryptocurrency.
- **C.2** advance data visualization—while regular user view the presented data in a typical format, advanced user may require more sophisticated visualization in order to gain more insight from the displayed result.

3.2.4 Wont Have

These are requirements which were agreed with the supervisor to be out of scope of the project. They could be implemented in a future stage

- **W.1** show user social network data. Social network can be useful to determine whether post about particular cryptocurrency have been coordinated or automated. another way to determine if there was a manipulation attempt is to track the source of information, but perhaps its more suitable for larger scale project.
- **W.2** Using unsupervised learning models. At the moment the focus of the project is to use different supervised machine learning models. if time permits, the project could use some of unsupervised learning models in order to see the performance difference.

4 Design

Figure 4.1 describe the workflow for this project. It starts from acquiring the relevant data, followed by data cleaning and feature engineering. Before the data inserted into the machine learning models, the data will be split into training, validation, and test set. Once the models were trained on the data, the resulting outcomes were presented.

4.1 Data Acquisition

As stated by Kamps and Kleinberg (2018), there is no centralized database that records pump and dump occurrences. Hence, the pump and dump history collected by Morgia et al. (2020) was utilised during this project. The collection consist of history of pump and dump that is organized in Telegram channel and Discord server dated as far back as July 2017 up until January 2021. The collection also gathered pump and dump event from five different cryptocurrency exchanges involving with more than 100 crypto coins. The project will solely concentrate on pump-and-dump occurrences that took place on the Binance exchange for two main reasons: First Binance is one of the largest cryptocurrency exchange. The second one is Binance provided comprehensive documentation of application programming interface (API) that allow retrieval of historical price data in minute interval.

Following that, a random selection of 100 pump and dump occurrences that took place on the Binance exchange was chosen. After the list of pump and dump events has been obtained, crypto coin market data in form of minute-level of open, high, low, close, and volume (OHLCV) was gathered from the queried crypto coin. The data was collected for a period ranging from 7 days before the pump start to 7 days after the pump event, resulting in a total of 15 days' worth of data which is subsequently saved in .csv files. It should be noted that minute-level OHLCV data gathered for the cryptocurrencies is based on Bitcoin price. The majority of cryptocurrency exchanges allow Bitcoin as a trading pair with other cryptocurrencies, which

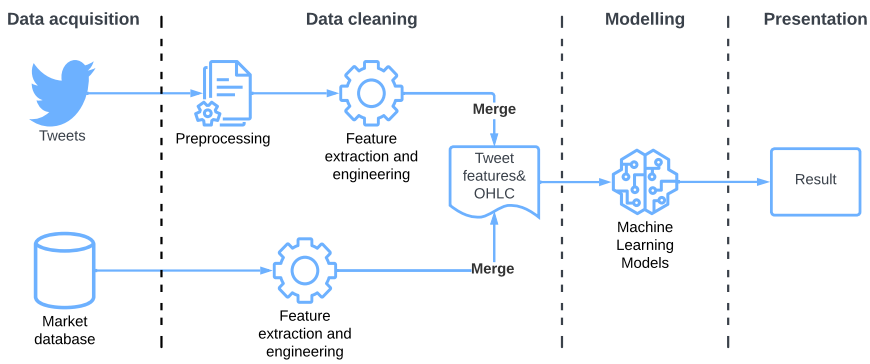


Figure 4.1: project workflow, starting from data acquisition up until presenting the result.

was done to maintain consistency for analysis. The following data comprising cryptocurrency information has been gathered and will be utilised for this project:

- **Timestamp**: The specific point in time where OHLCV data pertains to. It is often repented in in standardised format, such as UNIX time, which is number of seconds passed since 1st January 1970.
- **Ticker**: Cryptocurrencies' ticker (e.g., BTC).
- **Open**: The price of cryptocurrency at the beginning of time period.
- **Close**: Cryptocurrency price at the end of the time period.
- **High**: Highest price achieved by the cryptocurreny during time period.
- **Low**: Lowest price achieved by the cryptocurreny during time period.
- **Volume**: Volume of cryptocurrency traded during time period.
- **Pump mark**: An indication whether a cryptocurrency is being pump and dump during the time period. This is going to be the dependent variable for this project.

Originally, the collected cryptomarket data only has 6 out of 7 data from the list, it does not have Pump mark data. Pump mark data were added later by looking at Morgia et al. (2020) pump and dump history and then encode it the pump mark for each observation by following the convention:

$$\textit{Pump Mark} = \begin{cases} 0 & \text{coin is } \textbf{not} \text{ being pumped} \\ 1 & \text{coin is } \textbf{being} \text{ pumped} \end{cases}$$

Another challenging aspect is to determine the precise moment where the pump ends. In this paper, an end of a pump would be indicated by two ways: The first approach involves keeping an eye on the price and its changes. More specifically, it is considered proof that the pump has ended if the price significantly dropped. The second method involves analysing the volume data to determine when the pump stopped. If the pump continues for a period longer period of time and the price slightly fluctuates while the volume activity remains high, the end of the pump can be determined by looking at specific time frame where volume stared to normalized.

Additionally, tweets and other relevant social media data related to the coin that was written in English were also retrieved from Twitter by querying from the tweets based on the cashtag ($) from the same period, ranging from 7 days before the pump to 7 days after the pump had started. following data consisting of tweets information has been gathered and will be utilised for the purposes of this project:

- **Timestamp**: The specific point in time where the tweet are posted.
- **Username** and **userID**
- **Followers count**: could be useful for future analysis to investigate the social networking of pump and dump scheme
- **Verified status**: user account status, verified account means the user has significant amount of followers or paid $8 for Twitter blue.
- **Tweet**: The message or content of a tweet.
- **Tweet's engagement**: Number of likes, reply, retweet, and quote tweet
- **Hashtags** and **Cashtags**
- **URL**: link to the tweet
- **Pump and Dump status**: indicator whether a tweet related to pump and dump or not.

Originally, **Pump and Dump status** was made intended for use in a project aimed at detecting pump and dump schemes through the analysis of social signals and market data, which would be mapped crypto and social data to term frequency matrix. However, due to feasibility concerns, the concept was scrapped.

4.2 Data Cleaning and Feature Engineering

Following the collection of both market and social data, subsequent stage involves data cleaning, this includes pre-processing of tweet data, applying sentiment analysis in order to obtain the tweet's sentiment, followed by feature engineering for both market and social data and after that, both data are combined together by merging.

4.2.1 Data Cleaning

Text pre-processing was done in preparation to extract more obscure features from the tweet such as the sentiment of the tweet. The first step in tweet pre-processing started with removing URLs, mentions, hashtags and cashtags. Stop words—words that are insignificant and punctuation were further removed and the remaining text was converted into lowercase. The tweets were tokenized, which requires partitioning the tweets into individual word tokens. This was done to prepare the data for feature extraction and further analysis. After the tweets had been turned into tokenized words, the final step in pre-processing is stemming each word token.

Stemming is an algorithm that reduces all words which share a root, to a common form, generally, this is achieved by removing each word of its derivational and inflectional suffixes (Lovins 1968). For example, the word "cats" if applied to the stemming algorithm, would return "cat". There are many variations of stemming algorithms, but the most well-known algorithm used to stem English words is Porter's Algorithm (Porter 1980). The complete algorithm would take too long to explain, but the basic principles will be given. Porter's algorithm consists of 5 stages of word reduction and it is applied sequentially. There are several established conventions for choosing rules within each phase, such as choosing the rule from each ruling group that pertains to the longest suffix. This convention is applied to the following rule group in the initial step:

Rules	Examples
sses → ss	caresses → caress
ies → i	ponies → poni
ss → ss	caress → caress
s →	cats → cat

Later rules uses a concept known as *measure* of a word, which determines whether a word is long enough so that it is reasonable to consider the matching part of a rule as a suffix instead of being part of the word's stem by counting the number of syllables. For instance, the rule:

$$(m > 1) \quad EMENT \rightarrow \tag{4.1}$$

where m is equal to the number of letter before the phrase "*EMENT*". This means the word *replacement* would map to *replac*, but not *cement* to *c*.

4.2.2 Sentiment Analysis

Sentiment analysis is a branch of Natural Language Processing (NLP) which is focused with the identification of opinion from textual information (Indurkhya and Damerau 2010). Liu (2012) Defined sentiment analysis as ""the field of study that analyses people's opinions, sentiments, evaluations, appraisals, attitudes, and emotions towards entities such as products, services, organizations, individuals, issues, events, topics, and their attributes". As a way to analyse public's opinion on certain matter, features such as sentiment polarity and subjectivity can be examine.

Text polarity is one of feature that can be extracted from the text. Generally, sentiment polarity divided into two outcomes: **positive** and **negative**. However polarity can also be thought as a spectrum, generally spanning from -1 to 1 (Mejova 2014). a text with polarity of -1 implied

of highly negative sentiment towards a certain topic, whereas text with polarity of 1 signifies strong positive sentiment around a particular subject.

Beside polarity of a text, subjectivity could also helped to understand a person's sentiment towards a particular subject. Knowing the subjectivity of a text can be useful in understanding people's perspective and their stance with the subject matter. Subjectivity could also indicate person's bias towards certain topic that may be present in the text.

4.2.3 Feature Engineering

During this stage of data cleaning, extracting features from the gathered tweets is initiated. For this project, an effort was made to develop a set of features derived from both the current time-point data, such as the number of tweets posted and the total number of likes in a particular minute, as well as the recent past data, such as the average sentiment in the past x minutes. Following are the reported final tweet features used in this project:

- **Tweet**: The number of tweet posted at particular minute.
- **Tweet's engagement**: Number of likes, reply, retweet, and quote tweet at particular minute.
- **Polarity**: Polarity of a tweet or average polarity from multiple tweet at particular minute.
- **rolling avg polarity** and **rolling std polarity**: moving average and standard deviation of tweet's polarity over specific moving window.
- **rolling engagement**: moving sum of various engagement metric, such as number of likes, reply, retweet, and quote tweet over specific moving window.

For the cryptocurrency market data, several sets of features can be constructed without requiring any pre-processing steps. Features derived from crypto OHLCV data were developed in the same way as the tweet's feature constructed: by deriving both features from current time-point data as well as features that capture recent past data. Following are the reported final features from crypto OHLCV used in this project:

- **Timestamp**: The specific point in time where OHLCV data pertains to.
- **OHLCV**: Open, High, Low, Close, Volume data of crypto coin at particular minute.
- **Ticker**: Crypto coin symbol.
- **Pump mark**: Pump and dump indicator. 0 indicates there is no pump and dump activity at particular time, while 1 signal otherwise.
- **price change**: percentage change in price in respect to previous price data at particular minute. In this case, Price refer to Closed price in OHLCV.
- **rolling price change**: Sum of percentage change in price over specific moving window.
- **avg price** and **std price**: moving average and standard deviation of closing price over specific moving window.
- **avg vol** and **std vol**: Moving average and standard deviation of volume traded over specific moving window.

In order to determine the optimal configuration for our analysis, three separate features were created with different time window. In this project, durations of 5 minutes, 10 minutes, and 15 minutes were utilised for each set of features as the time window. After both features from crypto market data and tweet data has been extracted, crypto market data and tweet data were paired based the time window and timestamp. Therefore, we can assume the time series for each set of combined features as a vector:

$$M_f = [m_0, \ldots, m_{x-1}, m_x] \tag{4.2}$$

Where f represent the dimension of the features and x represent x minute, started from 7 days before the to 7 days after the pump.

4.3 Train-Test Splitting

Splitting dataset into training, validation, and test set was done with a ratio of 60%, 20%, 20% respectively In this project . Given that there are a total of 100 collected pump and dump files, it means 60 of these files will be allocated for the test set, while 20 files will be reserved for both the validation and test sets respectively. In order to achieve this, a partitioning was performed to separate train, validation, and test set. A subset of 20 files containing only one instance of a pump and dump event were randomly selected as a test set and put them in folder. while keeping the test and validation files in one folder.

Since the pump and dump event is considered to be a rare occurrence, the dataset is inherently imbalanced. An imbalanced dataset could lead to biased machine learning models and inaccurate predictions. The model might favour the majority class and the performance detection of the minority class may suffer. In the context of finding pump and dump, where a pump only last for no longer than an hour, if a machine learning trained with a biased dataset, it may result in a high number of false negatives, meaning the model might predict instances of a pump and dump as negative. In order to address the imbalanced nature of the dataset, a technique known as undersampling is chosen, which aims to reduce the proportion of majority class while keeping the minority class intact. One could argue undersampling crypto market data could make the model unable to detect a pattern for a pump and dump. However, because the features that were engineered included recent past data, the model is able to analyse the together with the current data together recent past data. With undersampling applied train and validation set size shrunk to just 1621 and 541 respectively. However, no undersampling was utilised in the construction of the test set. This is done to create realistic data representation from 7 days before the pump event occurring through 7 days after the pump had ended.

4.4 Naming Convention

Since array of models with different set of input features and classifiers were constructed, naming system is deemed necessary for ease of reference. models name such as logistic regression, support vector machine classifier, gaussian naive bayes, and random forest classifier will be referred as Logreg, SVC, GNB, and RFC respectively. In addition, each models also trained either only by market signal or social signal and market signal. The type of input is represented by the suffixes "*fin*" and "*fin + soc*" respectively. We labeled the fine-tuned models with a prefix "*ft*" to indicate their status. Lastly, a numerical value between 5, 10, 15 were added at the end referring to feature's window size in minutes. For example, Logreg *fin+soc* 5 would refer to a model that utilize default logistic regression using both 5 minute window time of market and social features.

4.5 Classifiers

During project, an array of machine learning models was constructed for pump and dump detection. The aim of constructing different machine learning models is to identify which model performed best than the rest of the models. For this project the use of supervised machine learning models are chosen. Supervised machine learning models is form of machine learning in which algorithm is trained on labelled dataset. Supervised machine learning works by providing a model input-output pair, where input can be considered as feature vectors and output can be treated as target value. The goal of supervised machine learning is to learn a function that able to map the feature vectors to their correct output value. Thus the algorithm is able to predict on unseen example (Russell and Norvig 2019). Four different machine learning models are used to find which models perform best. Following are the machine learning models used during this Project:

- **Logistic regression**

- **Support vector machine**
- **Naive Bayes**
- **Random Forest**

4.5.1 Logistic Regression

Logistic regression is statistical model used to analyse the relationship between dependent variable and multiple independent variable. Most common logistic regression model is a dichotomous outcome; it means logistic regression models only take two value as the predictor variables such as 0 or 1 using logistic function. Logistic function maps any input value to a probability between 0 or 1. The logistic regression function can be defined as:

$$y' = \frac{1}{1 + e^{-(\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \alpha)}} \tag{4.3}$$

where:

- y' is a predicted probability for target variable
- $(X_1, X_2 \cdots X_p)$ corresponds to combination of independent variables
- β are models learned weights and α is the model bias

α and β usually calculated with maximum likelihood estimates in which this method is preferred by Haberman (2014) and Schlesselman (1982).

Logistic regression can be applied wide area of expertise, Issitt et al. (2022) concluded logistic regression model approaches provided a reasonable performance in healthcare space Gong and Sun (2009); Ali et al. (2018); Ananthakumar and Sarkar (2017) used logistic regression to predict the stock market price movement and showed a degree of success.

For this project, 6 different logistic regression models were made following are the list logistic regression models and its configurations:

- Logreg *fin+soc* 5
- Logreg *fin+soc* 10
- Logreg *fin+soc* 15
- Logreg *fin* 5
- Logreg *fin* 10
- Logreg *fin* 15

This model differs in type of data received and the feature's window size. Every logistic regression are used with their default settings, except for *random_state* parameter. A number assigned to random state parameter in order to be able to reproduce deterministic outcome.

4.5.2 Support Vector Classifier

Support Vector Classifier (SVC) was another model leveraged for the purpose of the project. SVC is a supervised machine learning algorithm that works by creating hyperplane in N-dimensional space, where **N** is the set of features. The underlying objective of the algorithm is to establish a hyperplane that separates different classes by choosing data points or known as support vectors to maximize margin between classes. Subsequently, the hyperplane generated is employed as a decision boundary, allowing for the accurate classification of any new and previously unseen data points. Ahn et al. (2011) managed to construct SVC model that can be used as early warning system for detecting financial crisis. Hu et al. (2013) also demonstrated SVC capabilities to predict stock performance.

During the project, LinearSVC is used as our SVC models. The difference between LinearSVC and other SVC lies in model ability to draw the hyperplane. As the name suggest, LinearSVC generate decision boundary in straight hyperplane. Following are the list LinearSVC and its configurations:

- SVC *fin+soc* 5
- SVC *fin+soc* 10
- SVC *fin+soc* 15
- SVC *fin* 5
- SVC *fin* 10
- SVC *fin* 15

4.5.3 Naive Bayes

Another supervised model used for this project is **Naive Bayes**. A Naive Bayes models is one of probabilistic machine learning model that utilize Bayes theorem in order to predict classification. The term "Naive" indicates that the predictions count every features as independent, which means one feature does not affect the other.Naive Bayes can be define simply as:

$$P(A|B) = \frac{P(A)P(B|A)}{P(B)} \tag{4.4}$$

Using Naive Bayes, we can find the probabilities of **A** happening given **B** has occurred. When applied to classification problem, Naive Bayes can be expanded to:

$$P(y|x_1, \cdots, x_n) = \frac{P(y)P(x_1, \cdots, x_n|y)}{P(x_1, \cdots, x_n)} \tag{4.5}$$

where $(x_1, \cdots, x_n)$ represent features and y is target variable. We can simplify the equation to this:

$$P(y|x_1, \cdots, x_n) = \frac{P(y)\prod_{i=1}^{n} P(x_i|y)}{P(x_1, \cdots, x_n)} \tag{4.6}$$

Since $P(x_1, \cdots, x_n)$ a constant, therefore a proportionality is introduced:

$$P(y|x_1, \cdots, x_n) \propto P(y)\prod_{i=1}^{n} P(x_i|y) \tag{4.7}$$

$$\hat{y} = \underset{y}{\operatorname{argmax}}\, P(y)\prod_{i=1}^{n} P(x_i|y) \tag{4.8}$$

Using the above function will obtain the target variable, given set of features. Maximum A Posteriori (MAP) estimation was utilised to in order to estimate $P(y)$ and $P(x_i|y)$ where $P(y)$ is the relative frequency of class y in training set (sci).

several works have highlighted Naive Bayes capabilities in classification problem, Dey et al. (2016) demonstrated the ability of Naive Bayes to outperform K-Nearest Neighbour algorithm for classifying movie reviews with model accuracy achieved 80%. Goel et al. (2016) Also used Naive Bayes to predict twitter sentiment in real-time in tandem with SentiWordNet. In medical field, Naive Bayes demonstrated its capability to be able to predict heart disease given medical data with accuracy of 86.4198% (Vembandasamy et al. 2015).

For this project, one of Naive Bayes classifier is used, that is Gaussian Naive Bayes. Gaussian Naive Bayes assume features are independent and have Gaussian distribution. For classification task, Gaussian Naive Bayes calculate the probability of target variable by applying Bayes theorem

and using Gaussian probability density function to estimate the target variable given the features. Six different Gaussian Naive Bayes models were made following are the list Gaussian Naive Bayes and its configurations:

- GNB *fin+soc* 5
- GNB *fin+soc* 10
- GNB *fin+soc* 15
- GNB *fin* 5
- GNB *fin* 10
- GNB *fin* 15

4.5.4 Random Forest Classifier

Lastly, **random forest classifier** (RFC) is used for this project. random forest constructed by decision tress where each trees is generated using random vector sampled independently from the input vector, and each tree casts a unit vote for the most popular class to classify an input vector (Breiman 1999). random forest used ensemble approach—that is it utilize different models that worked together in order to make prediction, its called bagging. The idea of using bagging is to improve the accuracy of the random forest (Breiman 1996). random forest has proven its ability for classification task. For example, Morgia et al. (2020) managed to create random forest models that able to detect pump and dump in real-time in just five minutes and Azar et al. (2014) succesfully combine generic algorithm and random forest to detect lymph nodes with 92.2% accuracy.

In this project 6 random forest models were made, following are the list RFC and its configurations:

- RFC *fin+soc* 5
- RFC *fin+soc* 10
- RFC *fin+soc* 15
- RFC *fin* 5
- RFC *fin* 10
- RFC *fin* 15

5 Implementation

This section will discuss the implementation for acquiring the market data and the tweet data through the utilization from the chosen python package.

5.1 Finnhub

Finnhub is a financial data platform that offers free real-time stock prices. The platfrom offered market data from stock exchanges, 10 forex brokers, and more than 15 crypto exchanges. The platform also provides wide range of financial data ranging from market data, financial statements, list of shareholder ownership, to US bonds. Finnhub is a commonly used platform for gathering cryptocurrency market data due to its provision of minute-by-minute data at no cost. Moreover, Finnhub API documentation is user-friendly and straightforward to navigate.

After installing the Finnhub library, we were able to retrieve minute-level data for selected cryptocurrencies with ease using Finnhub's API. Down here is the implementation for fetching crypto coin OHLCV data from Finnhub.

```
import finnhub

api_key = "API_KEY"
finnhub_client = finnhub.Client(api_key)

# creating range of dates, interval=12 hours
# append the date to list and convert them into unix timestamp
date = [list of timestamp]
df = []

for i in range(len(date)):
   if date[i] == date[-1]:
      break
   dataBatch = (finnhub_client.crypto_candles(
      f'BINANCE:{token.upper()}{vs_currency.upper()}', '1', {date[i]},
          {date[i+1]-60}))
   loop_df = pd.DataFrame(dataBatch)
   df.append(loop_df)
   time.sleep(0.33)

concatinated_df = pd.concat(df, ignore_index=True)
concatinated_df.drop(columns='s', axis=1, inplace=True)
concatinated_df.rename(columns={'c': 'close', 'h': 'high', 'l': 'low',
                  'o': 'open', 't': 'timestamp', 'v': 'volume'}, inplace=True)
concatinated_df['ticker'] = token.upper()
concatinated_df = concatinated_df[['ticker', 'timestamp', 'open',
                           'high', 'low', 'close', 'volume']]

 # save the data into .csv
```

Listing 5.1: Fetching crypto data from Finnhub

Listing 5.1 Shows the implementation of utilising Finnhub API In order to collect the crypto coin OHLCV. In order to solve Finnhub API limitation, instead of parsing the whole date range, we opted to split the date range with interval of 12 hours and store them in **'date'** list. Then we implement for loop that iterates over the range of dates from **'date'** list. With each iteration, the code checks if current date is equal to the last date in the list. if so the code will stop iterating using **break** statement.

Assuming we are not in the final date, the code procced to retrieve the queried crypto coin data for the specific date time range. `Finnhub_client.crypto_candles()` method is used in order to fetch the queried crypto coin from Binance exchange with help of Finnhub API and stored it temporary in Pandas DataFrame object. Not to mention, the code enforce 0.33 second delay, This is done in order to resolve Finnhub 30 API calls/second limitation on top of free plan API limitation.

The stored DataFrame object then appended to **df** list. after all data had been retrieved, the code then proceed to concatenate the smaller DataFrame into one whole DataFrame that stores OHLCV data for 15 days. After that modification of the DataFrame was taken in order to easy access for the feature extracting process.

5.2 SNScrape

SNScrape is open-source web scrapped that focus on scrapping data from social networking service, hence the name SNScrape. it was developed in python and able scrap things such as user profile, hashtags, or even do queried search (e.g., finding relevant post). The reason why SNScrape was selected as the tool of choice during this project is because we want to find past tweets that could related to pump and dump event. Twitter limits its API usage on free developer account. However, Twitter accept an application for a request for more elevated Twitter data but it may take sometimes before the application gets accepted. Therefore, SNScrape was selected. As per 24th March 2023, SNScrape supports scrapping in different social media and here is the list of social media that can be scrapped using SNScrape:

- **Facebook**: user profiles, groups, communities (visitor's post)
- **Twitter**: users, user profiles, hashtags, queried searches, tweets, list posts, communities, and trends
- **Instagram**: user profiles, hashtags, and locations
- **Reddit**: users, subreddits, and searches (via Pushshift)
- **Telegram**: channels
- **VKontakte**: user profiles
- **Mastodon**: user profiles and toots
- **Weibo**: user profiles

Following, is the implementation of SNScrape in order to fetch tweet data:

```
import snscrape.modules.twitter as sntwitter

   tweets = []
   # collect the tweets
   for tweet in tqdm(sntwitter.TwitterSearchScraper(query).get_items()):
```

```
    tweets.append([tweet.date, tweet.user.username, tweet.user.id,
        tweet.user.renderedDescription, tweet.user.followersCount,
        tweet.user.verified,
               tweet.rawContent, tweet.replyCount, tweet.likeCount,
                   tweet.retweetCount, tweet.quoteCount, tweet.retweetedTweet,
                   tweet.quotedTweet, tweet.hashtags, tweet.cashtags,
                   tweet.url])

# create the dataframe
df = pd.DataFrame(tweets, columns=['Time', 'Username', 'ID', 'Description',
    'FollowersCount', 'Verified', 'Tweet', 'ReplyCount', 'LikeCount',
                           'RetweetCount', 'QuoteCount', 'Retweet',
                               'QuoteTweet', 'Hashtags', 'Cashtags', 'URL'])

# remove timezone converter
df['Time'] = pd.to_datetime(df['Time'].dt.tz_convert(None))
df = df.sort_values('Time')

#save the tweet into a dataframe
```

Listing 5.2: *Fetching tweets data from Twitter by utilizing SNScrape.*

Listing 5.2 display the implementation of using SNScrape to collect tweets. First, we initiated an empty list called **tweet**. after that it runs a loop that iterate over the collected tweets collected by `sntwitter.TwitterSearchScraper(query).get_items()`. it collected sevreal information such as timestamp, username, userID, follower count, etc. after all tweets had been collected, the code creates Pandas DataFrame object and stores the tweet inside the DataFrame. The DataFrame has columns corresponding to each tweet data collected. Finally the code modified '**Time**' column to Pandas DateTime object and removes timezone information and finally sorted the tweet in ascending order.

6 Evaluation

In this part, we will discussed the models performance on different setup. The first task to evaluate each models on validation set to find which model perform best. After we found the best model, the selected models will undergone fine-tuning process in order to get better result. Using the fine-tuned model then we try compare its performance with the rest of the model on test set and to see the accuracy to predict a pump and dump event started from 10 minutes before the pump until 10 minutes after the pump had ended.

6.1 Evaluation Metric

To assess different model setups on their ability to predicting pump and dump event , we use precision , recall, and f1 score, as these three metrics are widely used to see the models performance. During comparing different classifiers, we tried to highlight the performance of the classifier when making predictions for class 1, that is when the pump is happening in validation and test set. Later on, when we are analysing the test set, we will also provide the confusion matrix to provide a better understanding.

Precision described as the rate of which our model is able to predict the true positive out of total positives. Mathematically, precision can be defined as:

$$\text{recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}} \tag{6.1}$$

Recall, also known as true positive rate is a measure of predicted positives out of the true positives. Mathematically, recall can be describe as:

$$\text{precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}} \tag{6.2}$$

F-1 score is a harmonic mean mean of precision and recall and it provides balanced measure of the model overall's performance. F-1 score is useful when the dataset used is imbalanced, as it take both precision and recall. Mathematically F-1 score can be describe as:

$$F1 \text{ score} = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} \tag{6.3}$$

For the second task we will use accuracy as a measure to compare between fined-tuned models in specific time range, that is 10 minutes before the pump until 10 minutes after the pump ended. Accuracy can be described as the number of correct guesses out of total guesses made, mathematically it can be written like this:

$$\text{accuracy} = \frac{\text{correct prediction}}{\text{total prediction}} \tag{6.4}$$

6.2 Validation set Result and Fine-Tuning

6.2.1 Comparing Classifiers on Validation Set

Table 6.1 present the performance matrix for all of our models on validation set. Models are grouped based on its architecture features used, with those utilizing shorter window time first.

For logreg models, all of them produced very high recall value of 1, which means all logreg model is able to correctly identified all true positives. However their precision and f-1 score is quite low. Surprisingly, both logreg *fin+soc* 10 and logreg *fin* 10 is having similar result, means that different features and the time window does not affect the models performance. This also can be seen in logreg *fin+soc* 15 and logreg *fin* 15 as well. While the precision of logreg *fin* 5 gained by 0.32 from logreg *fin+soc* 5.

For our SVC models, we observe SVC *fin+soc* 10 and SVC *fin+soc* 15 having similar value of precision score, but SVC *fin+soc* 15 has higher recall value. When comparing to SVC *fin+soc* to SVC *fin*, the difference of precision and recall between SVC *fin+soc* 5 and SVC *fin* 5 is quite significant, while SVC *fin+soc* 5 has better precision, SVC *fin* 5 has better recall score. But the most surprising thing is that SVC *fin* 15 inhibits the lowest recall and f-1 score with 0.244 and 0.385 respectively making it the worse preforming models out of 24 models.

Our Gaussian Naive Bayes also shares some interesting insight. They have the highest average of precision score and both GNB *fin+soc* 5 and GNB *fin* 5 have the same precision score regardless of features used. While having the highest precision value of 0.990, both GNB *fin+soc* 5 and GNB *fin* 5 have the second lowest recall score of 0.370. Both GNB *fin+soc* 5 and GNB *fin* 5 also have similar value f-1 score of 0.539. Similar score on different features also inhibits in GNB *fin+soc* 10 and GNB *fin* 10, where both model achieved recall of 0.796 and f-1 score of 0.878.

Lastly, our RFC models showed very promising performance result. RFC models managed to have have second highest average in both of precision and recall. Therefore, RFC models achieved the highest average f-1 score from three different models. In particular, RFC *fin* 15 is having the highest f-1 score of 0.976 while also pertaining recall value of 0.978. There are no RFC models that have similar value in recall, precision and f-1 score, however RFC *fin+soc* 5 attained close value of the three metrics score to RFC *fin* 5 with precision, recall, and f-1 score of 0.952, 0.963, and 0.958 respectively. Even though RFC *fin+soc* and RFC *fin* 10 also share little differences in precision, recall, and f-1 score, the difference is not as close when between comparing RFC *fin+soc* 5 and RFC *fin* 5.

6.2.2 Validation Set Analysis

In conclusion, we can observe that each model have their own strength and weaknesses in predicting pump and dump activity in validation set. While logreg models perform well in terms of recall, they have average precision and f-1 score, meaning that logreg models tend to classifiy more false positives despite high recall score. SVC models have mixed performance, while some models having high value of precision, recall, and f-1 score, there is also SVC models that performed worse with lower recall and f-1 score. For GNB models, while all of the models is having highest precision score, they also have lower recall value. It means GNB models is tend to classify more false negative. Lastly, our RFC models showed to be the best model out of three other models. While not having highest recall value like logreg model or having the highest precision value like GNB models, they still keep a high number of precision and recall. Hence, that is why RFC model is having the highest average f-1 score.

Models	Validation set		
	precision	recall	f-1 score
logreg *fin+soc* 5	0.522	**1**	0.711
logreg *fin+soc*10	0.601	**1**	0.751
logreg *fin+soc* 15	0.507	**1**	0.672
logreg *fin* 5	0.554	**1**	0.713
logreg *fin* 10	0.601	**1**	0.751
logreg *fin* 15	0.507	**1**	0.672
SVC *fin+soc* 5	0.980	0.719	0.829
SVC *fin+soc* 10	0.957	0.896	0.925
SVC *fin+soc* 15	0.942	0.837	0.886
SVC *fin* 5	0.704	0.907	0.793
SVC *fin* 10	0.959	0.856	0.904
SVC *fin* 15	0.904	**0.244**	**0.385**
GNB *fin+soc* 5	**0.990**	0.370	0.539
GNB *fin+soc* 10	0.977	0.796	0.878
GNB *fin+soc* 15	0.983	0.648	0.781
GNB *fin* 5	**0.990**	0.370	0.539
GNB *fin* 10	0.977	0.796	0.878
GNB *fin* 15	0.983	0.652	0.784
RFC *fin+soc* 5	0.952	0.963	0.958
RFC *fin+soc* 10	0.970	0.967	0.968
RFC *fin+soc* 15	0.967	**0.978**	**0.972**
RFC *fin* 5	0.953	0.970	0.961
RFC *fin* 10	0.964	**0.978**	0.971
RFC *fin* 15	0.974	**0.978**	**0.976**

Table 6.1: *The performance of the classifier when making predictions for class 1 (when pump and dump is occurring) in validation set.*

6.2.3 Fine-Tuning Random Forest Classifier

Before we proceed to analyse our models on test set, we decided to fine-tune RFC models in hope that the models will perform better. For that we use GridSearchCV, an exhaustive search over a specified parameter grid. The hyperparameters being tuned for the RFC model include the number of estimators, the maximum depth of the tree, the minimum number of samples required to split an internal node, the maximum number of features to consider when looking for the best split, the maximum number of leaf nodes in the tree, the minimum number of samples required to be at a leaf node, and whether or not to use bootstrapping when building trees. And then the RFC model will be trained and evaluated with each combination of hyperparameters three times using cross-validation. Finally, the GridSearchCV will list the set of parameters that will yield the best result.

6.3 Comparing Classifiers on Test Set

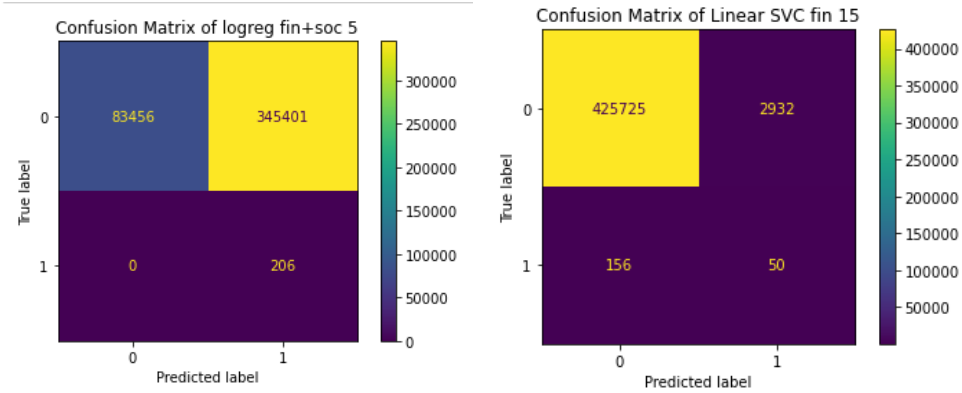
Looking at Figure 6.2, this time we can identify that both precision and f-1 score have been dropped significantly. This is due to the nature of imbalanced dataset. For the test set, logreg is still able to retain its recall value from validation set. However, all logreg models seems to have same precision score of 0.001, which is the lowest precision score out of three models. This means on test set, the logreg model will often make false positive predictions. For the f-1 score, all six models have closely similar value. Moving into SVC, this models also dropped in terms of precision score, although its not as severe as logreg models. SVC *fin+soc* 5 achieved the highest precision and f-1 score when compared to another SVC models. Both SVC *fin+soc* 10 and SVC *fin+soc* 15 have same precision score and similar value of recall. Surprisingly, SVC *fin* 5 have the same precision value from the logreg models.Tthe lowest recall out of different models can be found on SVC *fin* 15

For our GNB models, they have the highest precision value compared to another three classifier. Again GNB *fin+soc* 5 and GNB *fin* 5 possed th highest precision score among another models with precision of 0.132. Both models also share close f1-score with GNB *fin+soc* 5 f-1 score reached 0.179 while GNB *fin* 5 f-1 score peaked at 0.180, the highest f1 value compared to another models. despite having the highest precision and f-1 score the issue from GNB model being low in recall still persist. Lastly, our RFC models also seems have lower precision score. However, our RFC models is still retain its high recall score. Both RFC *fin+soc* 5 and RFC *fin* 10 have slight difference in the precision, recall and f-1 score. with the difference of 0.001, 0.005, and 0.002 in precision, recall and f-1 score respectively. RFC *fin* 15 produced the highest recall when compared to another model, with recall value of 0.971, 0.15 difference from RFC *fin+soc* 15.

In general, all model's performance dropped significantly when applied to the test set. This is because the imbalanced nature of our test set that represent real world pump and dump data, where pump and dump event can only be found short period of time. While logreg models retain its recall capability, its precision dropped significantly. Most of the SVC models performed moderately well. While some models acheiving higher presicion and f-1 score than others. Moving on to GNB models, the same issue on validation set still persist. Despite its precision dropped due to imbalanced dataset, GNB model still ability to retain high precision value compared to other models. Lastly, our RFC models is still able to retain high recall value, despite being tested in imbalanced dataset. Because of that, the RFC model precision also dropped, even lower than GNB models.

Models	Test set		
	precision	recall	f-1 score
logreg *fin+soc* 5	0.001	**1**	0.001
logreg *fin+soc*10	0.001	**1**	0.002
logreg *fin+soc* 15	0.001	**1**	0.001
logreg *fin* 5	0.001	**1**	0.001
logreg *fin* 10	0.001	**1**	0.002
logreg *fin* 15	0.001	**1**	0.001
SVC *fin+soc* 5	0.049	0.689	0.091
SVC *fin+soc* 10	0.021	0.864	0.041
SVC *fin+soc* 15	0.021	0.859	0.040
SVC *fin* 5	0.001	0.869	0.003
SVC *fin* 10	0.023	0.864	0.045
SVC *fin* 15	0.017	0.243	0.031
GNB *fin+soc* 5	**0.132**	0.282	**0.179**
GNB *fin+soc* 10	0.053	0.714	0.098
GNB *fin+soc* 15	0.068	0.592	0.122
GNB *fin* 5	**0.132**	0.282	**0.180**
GNB *fin* 10	0.053	0.714	0.098
GNB *fin* 15	0.069	0.597	0.124
ft RFC *fin+soc* 5	0.035	0.966	0.068
ft RFC *fin+soc* 10	0.035	0.961	0.068
ft RFC *fin+soc* 15	0.041	0.956	0.078
ft RFC *fin* 5	0.036	0.961	0.070
ft RFC *fin* 10	0.034	0.961	0.066
ft RFC *fin* 15	0.042	**0.971**	**0.081**

Table 6.2: *The performance of the classifier when making predictions for class 1 (when pump and dump is occurring) in test set.*



(a) Logreg fin+soc 5 confusion matrix *(b) SVC fin 15 confusion matrix*

Figure 6.1: *Confusion matrix for both logreg fin+soc 5 and SVC fin 15 respectively. Figure 6.1 shows logreg models produced high amount of false positive predictions, while majortiy of SVC model is able to predict the target class quite well, with the exception SVC fin 15 (figure 6.1b).*

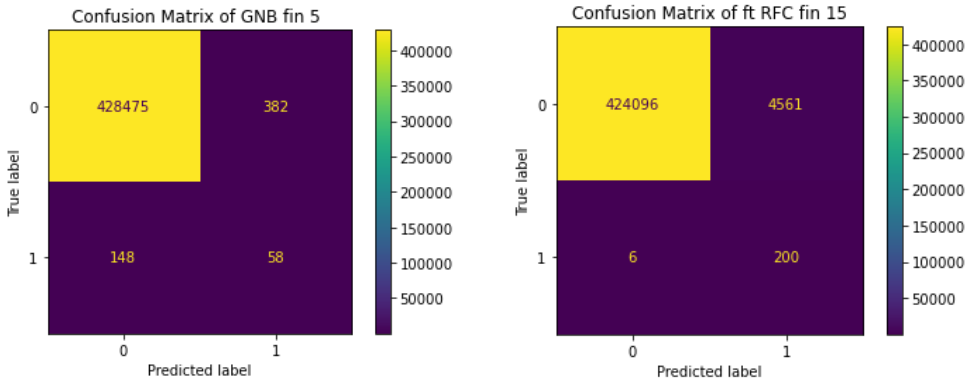
6.3.1 Test Set Analysis: Confusion Matrix

In binary classification, confusion matrix is used in order to provide better understanding where the models is able to perform well and where the model fails to identify the class correctly. For This project we want to emphasise on models ability to predict

For our logreg models, the model able to correctly predict all the true positives, in this case the time when pump and dump is occurring. However, the amount of false positive in this model where the problem starts. Figure 6.1a shows the example of confusion matrix from logreg *fin+soc* 5. With more than 300,000 datapoints falsely labelled, this model will predict false positives nearly in all occasion. Other logreg model also have similar confusion matrix, see *appendix* for more details. Our SVC model performs better than our previous model. SVC is able to predict majority class, in this case the time when the pump is not occurring most of the time. it also able to predict the time when the pump and dump started quite well. However not all SVC model is able to guess the target classes quite well. In this example SVC *fin* 15 (6.1b) is having large number of false negative. For the purpose of this project we do not want a model have high number of false negative. It means, when pump and dump is occurring, our model will most likely predict will predicted pump and dump is not happening. For other SVC confusion matrix, please look at *appendix*.

GNB confusion matrix has produced mixed result of confusion matrix, where one model having considerable amount of true positive correctly guess and other model is having high number of false negative. Figure 6.2a illustrate when the model prediction ended up with high number of false negative. On the other hand, our RFC models produce the most accurate prediction compared to other three classifiers. Figure 6.2b shows SVC *fin* 15 model is able to predict most of pump and dump instances while having considerable amount of false positives. This is true for other SVC models. For more confusion matrix details, please look at *appendix*.

In this project, we are focused on finding the instance where the pump and dump is happening in crypto market. After evaluating the models performance with precision, recall, and f-1 score, we use confusion matrix in which provide insights into the models' ability to correctly predict true positives and avoid false negatives. Based on the confusion matrix, Logreg models were able to predict all instances of pump and dump. However, the model also suffered from high degree of false positives. SVC models performed better than logreg models, even though some models produce high number of false negative and other model produce high number of false positives.



(a) GNB fin+soc 5 confusion matrix *(b) RFC fin 15 confusion matrix*

Figure 6.2: *Confusion matrix for both*

GNB models generated mixed results, with some model correctly predicting considerable amount of true positives while others experienced high number of false negative. Finally, our RFC models were found to be the best model compared to the rest of the three models. Most of RFC models were able to predict most pump and dump instances, while maintaining considerable amount of false positives. Overall, RFC models were found to be the best-performing models in this project.

6.4 Analysis: Feature Differences

6.4.1 Time Window Difference

When looking at Table 6.1 and Table 6.2, we can see the models performance based different sets of features. If we were to compare model performance by looking at utilized time window, we can observe logreg performance give us no obvious pattern on validation set. The result of precision, recall, and f-1 score is not showing any pattern as the window time increased. However on test set, logreg is able to retain precision, recall, and f-1 score. SVC produce quite interesting result on validation set, Thus theres no pattern to examine.

On SVC *fin+soc*, if we increase our window time on validation set, the precision value will decrease while the recall score will increase. On the other hand SVC *fin* experienced the opposite, the precision increase and recall will be decrease as we enlarge the time window. Because of this, both SVC *fin+soc* 10 and SVC *fin* 10 produced the highest f-1 score out of SVC models. On the test set, larger time window showed to decreasing SVC *fin+soc* f-1 score but not SVC *fin*. However, in test set, SVC models produce unusual pattern, where precision decrease on SVC *fin+soc* but on SVC *fin* precision will increase by 0.022 from SVC *fin*5 to SVC *fin*10 before it went down to 0.017 on SVC *fin*15.

When looking at GNB model results on validation set, larger time window seems to change the models precision, recall, and f-1 score in no clear pattern. GNB exhibit same pattern on both validation and test set, where its precision would drop when increasing time window GNB 5 to GNB 10 before precision increased on GNB 15 regardless of the feature selection.

Finally, when we looked at RFC models that used in validation set, we can observe that larger time window leads to higher f-1 score for both RFC *fin+soc* and RFC *fin*. Interestingly, as the time window grows larger, the recall on RFC *fin+soc* will also increase while its precision score will increase 0.018 from RFC *fin+soc* 5 to RFC *fin+soc* 10 and then dropped back to 0.967 on RFC *fin+soc* 15. The recall also increased from 0.970 to 0.978 on RFC *fin+soc* 10 and stays at RFC

*fin+soc*15. In the test set, RFC *fin+soc* get higher f-1 score as the time window increased, but its precision is also decreased. However, on RFC *fin* recall will increased as the time window gets bigger. While f-1 score will decrease at RFC *fin* 10 before reaching its highest value on RFC *fin* 15.

6.4.2 Analysis: Financial or Financial + Social Features

our models show to behave differently when different financial or social data is used for detecting pump and dump. On logreg models the is slight to none difference between using financial and social signals and only using financial signals, both on validation set or the test set. However SVC models, using both financial and social signal proved to be usefull in order to determining pump and dump event. Moving on to GNB models, both result from validation and test set show no changes in precision, recall, and f-1 scores regardless the difference in financial and social data, or just financial data. Finally from RFC models, financial data showed to be the best features for the models, averaging higher in recall, precision, and f-1 score compared to its counterpart. This holds true on both validation and test set. One possible explanation for why RFC models that use financial data yield the highest result, is due lack of social data gathered for each cryptocurrency. While OHLCV data follows minute-by-minute basis, a tweet published about certain cryptocurrencies do not have the same density. In other words, tweets are published irregularly and do not follow any specific interval. Hence, that is why RFC model that utilised both financial and social signal preformed worse.

6.4.3 Closing Remark

in conclusion, different set of features can affect the machine learning models differently. Some models might be able to draw a pattern for each features changed while some not. Moreover, the effect of changing the time window on model performance is also dependent on the specific model and feature selection. The findings emphasize the importance of careful feature selection and tuning to achieve optimal performance in machine learning tasks.

6.5 Accuracy and Finding Pump and Dump Event in Specific Time Range

Table 6.3: *Accuracy of different RFC models when predicting pump and dump event. Started from 10 minutes before the pump and ended 10 minutes after the pump ends.*

Models	Accuracy
RFC *fin+soc* 5	65.7%
RFC *fin+soc* 10	66.098%
RFC *fin+soc* 15	66.83%
RFC *fin* 5	65.978%
RFC *fin* 10	64.323%
RFC *fin* 15	66.027%

In this part we tried to calculate the fine-tuned RFC models accuracy and we tried to determine whether our fined-tuned RFC model is able to detect the beginning and the end of pump and dump event. Table 6.3 shows the accuracy of RFC models when detecting pump and dump during selected time period. All of RFC models have similar accuracy result averaging 65.801%. While it looks underwhelmed, we can investigate our machine model prediction by showing two different graph and gain some insight from it. Another graph can be look at the gitlab repository (link is attached in *appendix*).

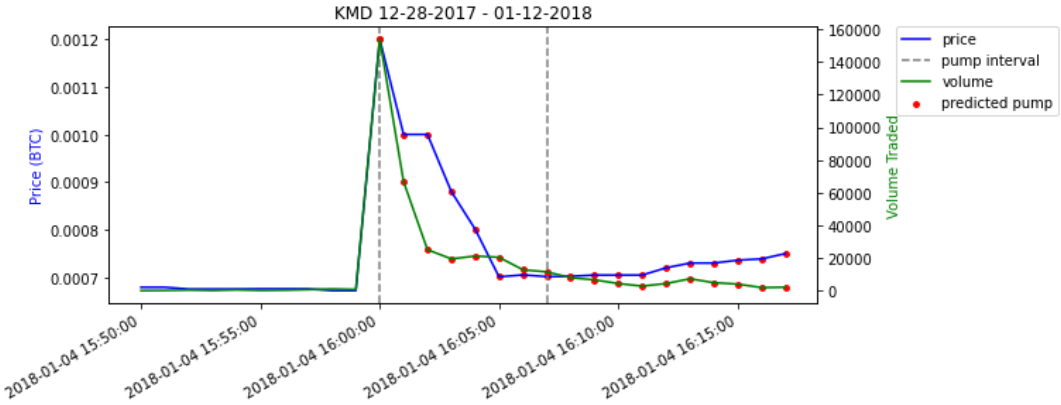


Figure 6.3: *Pump and Dump Event from* ***KMD*** *token. started at 16:00 UTC at 4th January 2018 and ended 7 minutes after.*

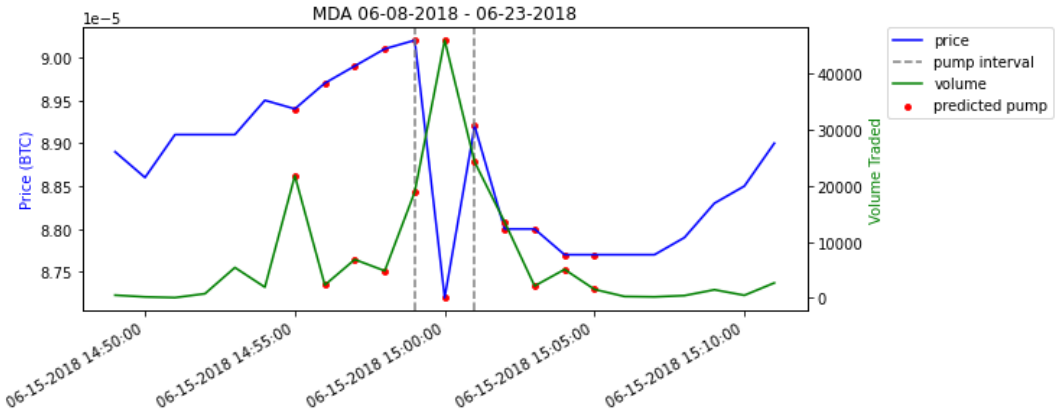


Figure 6.4: *Pump and Dump Event from* ***MDA*** *token. started at 14:59 UTC at 15th June 2018 and ended 2 minutes after.*

Figure 6.3 shows the models prediction during **KMD** token pump and dump event that carried out on 4th January 2018. In this graph, Price is indicated with blue line and price is measured with BTC because BTC as its widely used as cryptocurrency exchange. Volume trend is represented with the green line. Two grey dash line used as a separator between pump and dump event, the first grey dash line indicate the pump is started, and the second dash line indicates that the pump is over. Any data point lies within the dashed line indicate the price and volume movement during the pump. From this graph, we could assume that the model is able to predict the start of the pump and dump event, however, it fails to find the time where the pump ends.

Another example is taken from pump and dump event of **MDA** token. In this example, the models predict the start of the pump too early. It maybe undesirable to have models that predict pump start to early, but since the start of the pump is determined by Telegram or Discord admin, it could argued that minutes before pump and dump started the admins and VIPs is trying to get the coin in much lower price in order to gain more profit.

This is the main issue that arise from the model's predictive ability represents a significant challenge

that needs to be addressed. A better way to determined when the pump and dump event has ended is required in order to improve the model's accuracy. By finding and constructing better method to determine the end of pump and dump event, the models can generate more accurate predictions, which in turn also will help a lot of people by distinguishing time period where the pump is happening.

7 Conclusion

Cryptocurrencies Pump and dump schemes have become increasingly common due to the thier high volatility, their decentralised nature, and they are highly accessible to the general public. Furthermore, the current lack of regulation for cryptocurrency market makes it simple for dishonest individual or groups to exploit the cryptocurrency market for profit. Social media activities also help to contribute to prevalence of pump and dump scheme, by enabling gathering of individuals who plot to such scheme and by creating a sense of urgency in general public.

In this project, we attempted to devise a machine learning algorithm that takes both of social and market signal as our features. We managed to gather minute-data for both social and market data, then we attempted to construct a set of features that will be inserted to different type of supervised machine learning model. Finally we will compare each models with their set of features

From the experiment conducted, we found that Random Forest Classifier that only use financial signal with 15 minute time window has the best result from other other models. It has precision, recall and f-1 score of 0.974, 0.978, and 0.976 respectively in validation set. On the test set, the same models achieved precision, recall, and f-1 score of 0.042, 0.971. and 0.081 respectively.

We also conclude that different features behaves differently on different machine learning models. As why RFC models perform better when it only use financial data, it can be inferred that the social data gathered is not in same density as our financial data collected. Therefore, it could be acted as a noise in our datapoints.

Finally, for determining pump and dump on selected time interval. Our fined-tuned RFC model displayed considerable performance with six models averaging 65.801% accuracy. When we look deeper, the model fails to determine the end of the pump within the time interval Additionally, there are instances where the model is predicting the time of the pump to occur earlier that it actually does, which could cause outcomes to vary more significantly. In order to address this problem, A method to determine where the pump and dump event ended is needed in order to create better prediction.

7.1 Future Work

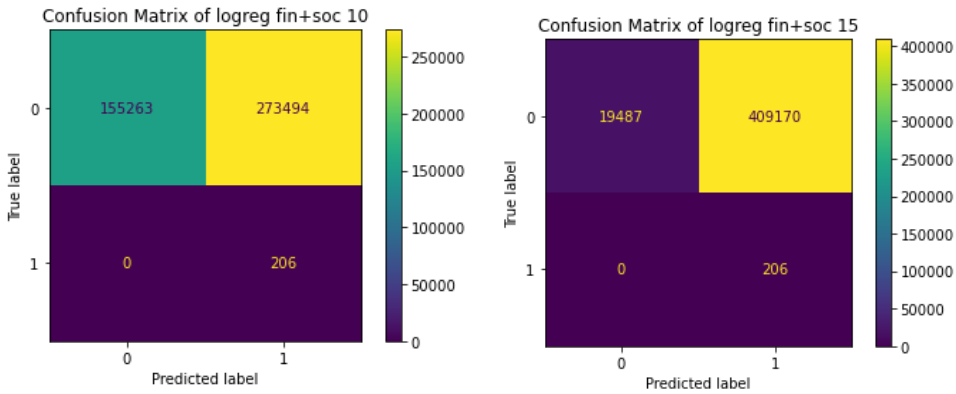
The subsections that follow list ideas for future work, should this endeavour be extended in the future

- **real-time analysis**: with the current implementation that only used historical data, we could implement the step taken in this project for real-time data analysis.
- **use the models on bigger cryptocurrencies**: For the future, also can apply the process taken in ths project for well-known cryptocurrencies, such as Bitcoin, Ethereum, Dogecoin, etc.
- **Web based application**:Instead of storing and displaying the result of analysis locally, future works could use a web application in order to gather the required data, processes it and the give the prediction back to the user.

- **Unsupervised learning model**: while this project only compares difference supervised machine learning model, the future works can expand this project by applying different unsupervise learning models such as long short term memory to compare with supervised machine learning models. (LSTM).
- **utilizing social netowrk data as features**: this is done to gain insight behind the social media posted by indiviuals. By looking at social data we could extract some precious information on how the tweet might be posted originated from the same person, finding if tweet posted are shared by bots, etc.

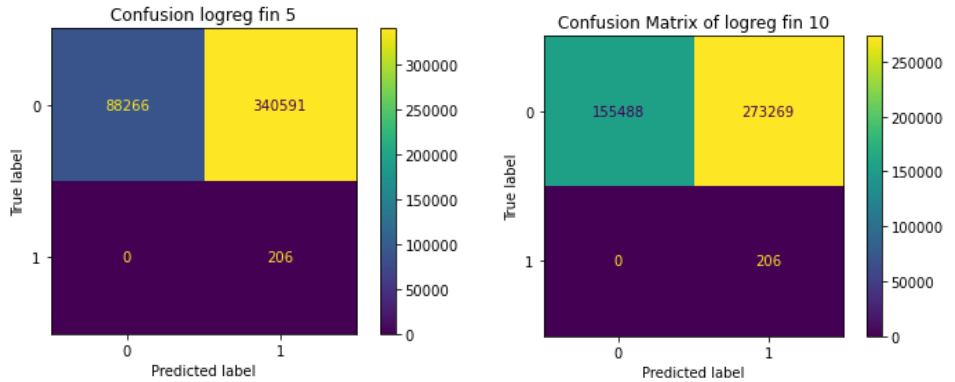
A Appendices

- **Gitlab** link: https://stgit.dcs.gla.ac.uk/2512262a/individual-project



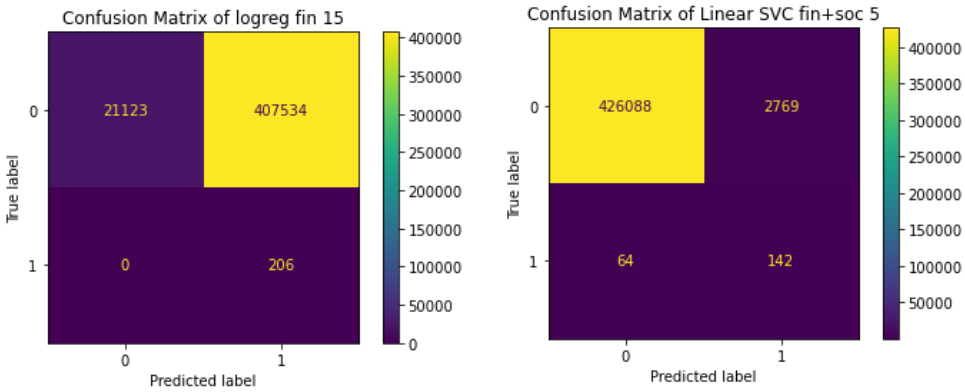
(a) Logreg fin+soc 10 confusion matrix *(b) Logreg fin+soc 15 confusion matrix*

Figure A.1: *Logreg fin+soc 10 confusion matrix and Logreg fin+soc 15 confusion matrix*



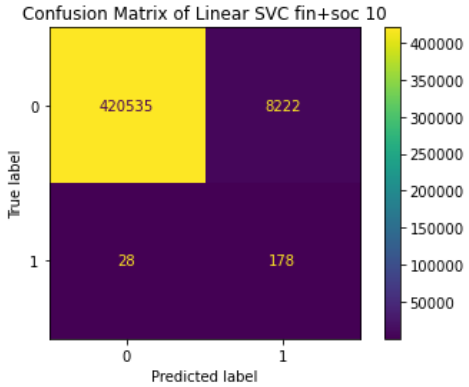
(a) Logreg fin 5 confusion matrix *(b) Logreg fin 10 confusion matrix*

Figure A.2: *Logreg fin 5 confusion matrix and Logreg fin 10 confusion matrix*

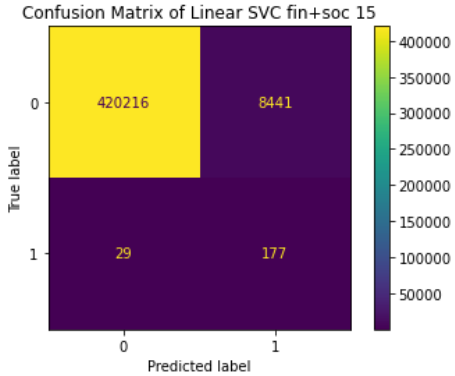


(a) Logreg fin 15 confusion matrix *(b) SVC fin+soc 5 confusion matrix*

Figure A.3: *Logreg fin 15 confusion matrix and SVC fin+soc 5 confusion matrix*

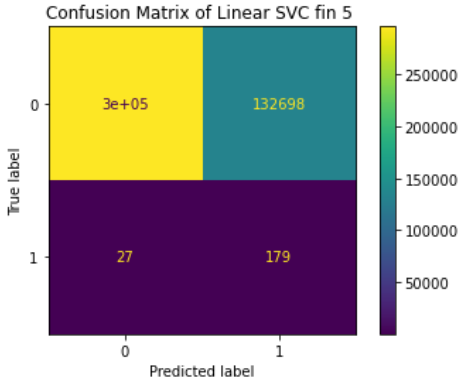


(a) SVC fin+soc 10 confusion matrix

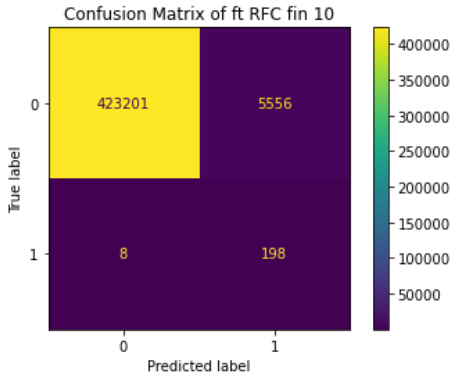


(b) SVC fin+soc 15 confusion matrix

Figure A.4: *SVC fin+soc 10 confusion matrix and SVC fin+soc 15 confusion matrix*

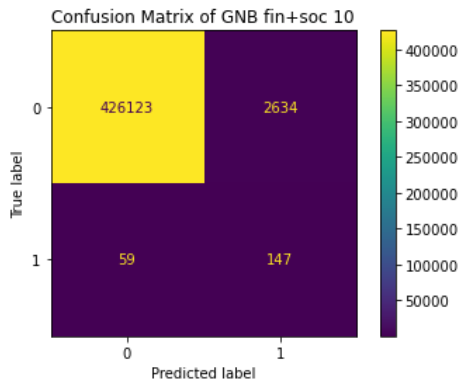


(a) SVC fin 10 confusion matrix

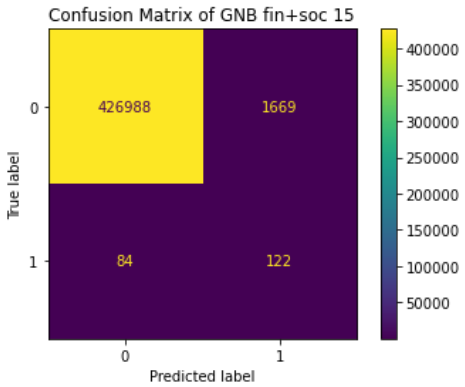


(b) RFC fin 10 confusion matrix

Figure A.5: *SVC fin 5 confusion matrix and RFC fin 10 confusion matrix*

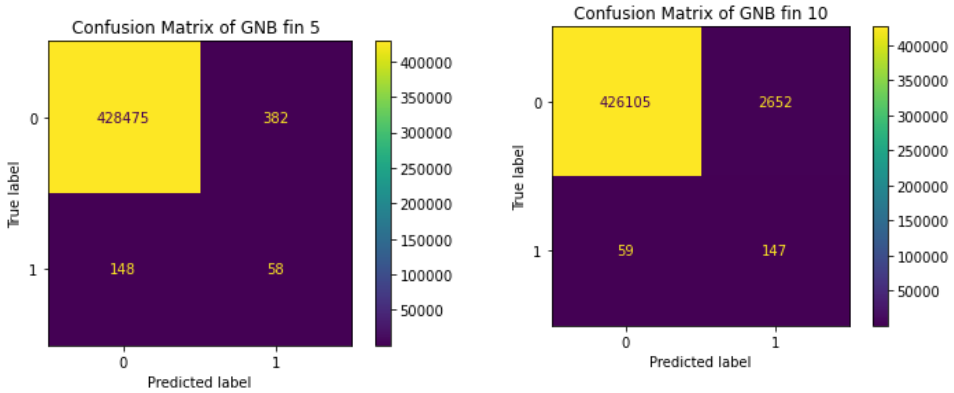


(a) GNB fin+soc 10 confusion matrix



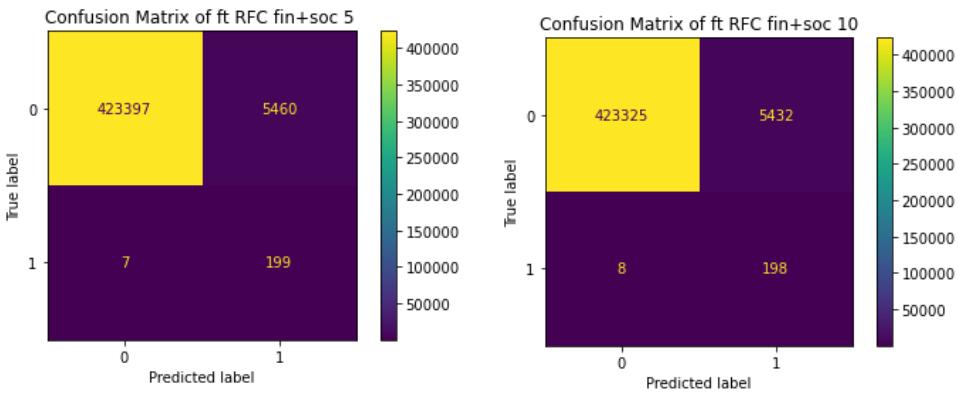
(b) GNB fin+soc 15 confusion matrix

Figure A.6: *GNB fin+soc 10 confusion matrix and GNB fin+soc 15 confusion matrix*



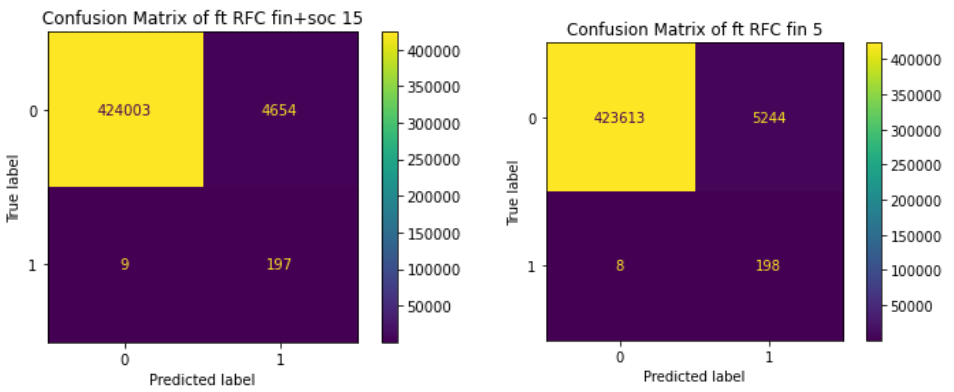
(a) GNB fin 10 confusion matrix *(b) GNB fin 15 confusion matrix*

Figure A.7: *GNB fin 10 confusion matrix and GNB fin 15 confusion matrix*



(a) RFC fin+soc 5 confusion matrix *(b) RFC fin+soc 10 confusion matrix*

Figure A.8: *RFC fin+soc 5 confusion matrix and RFC fin+soc 10 confusion matrix*



(a) RFC fin+soc 15 confusion matrix *(b) RFC fin 5 confusion matrix*

Figure A.9: *RFC fin+soc 15 confusion matrix RFC fin 5 confusion matrix*

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